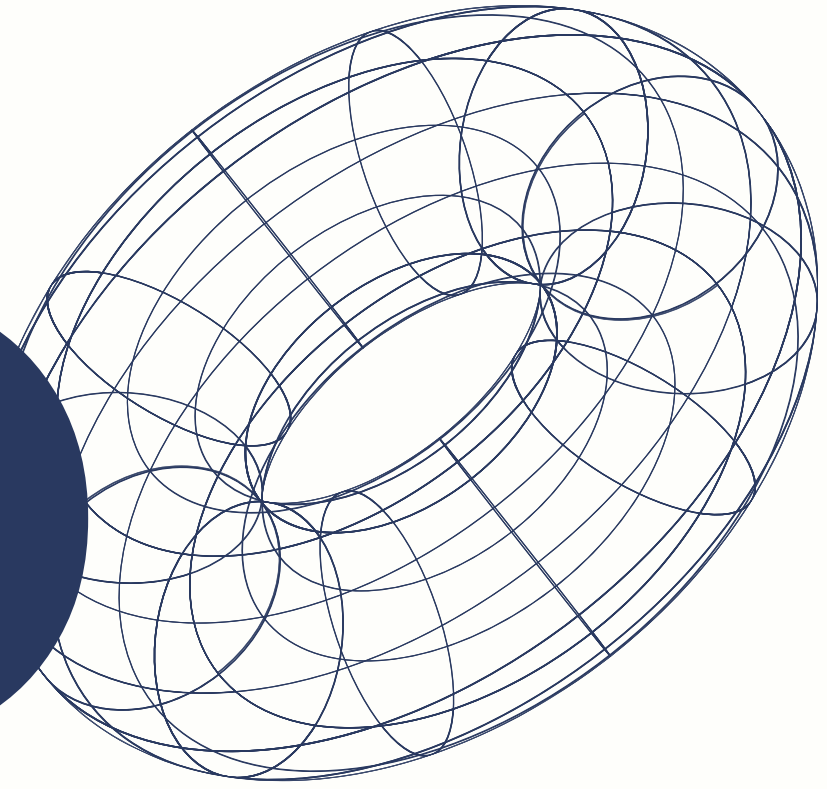


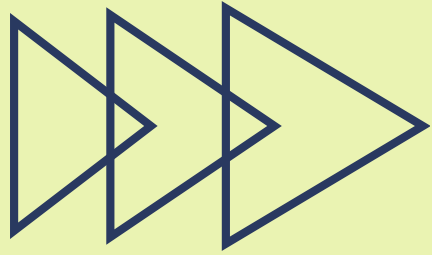
Vietnamese-German University

year 2026

group project



by: AE CÂY KHẾ TEAM



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project overview.

#groupproject

Indoor plants require regular monitoring and watering.

However:

- Users may forget to water plants.
- Overwatering can damage roots.
- Environmental conditions affect plant health.

Proposed Solution

A Smart Plant Monitoring and Automated Watering System using:

- ESP32 microcontroller
- Soil moisture sensor
- DHT11 temperature and humidity sensor
- Mini water pump
- IoT monitoring platform

The system automatically waters plants when soil moisture becomes too low.

/page 02

group members

#groupproject

10423193-- Lê Trung Đức

10423195-- Võ Hồng Phúc

10423196-- Trần Văn Hưng Thịnh

project goals.

/page 03

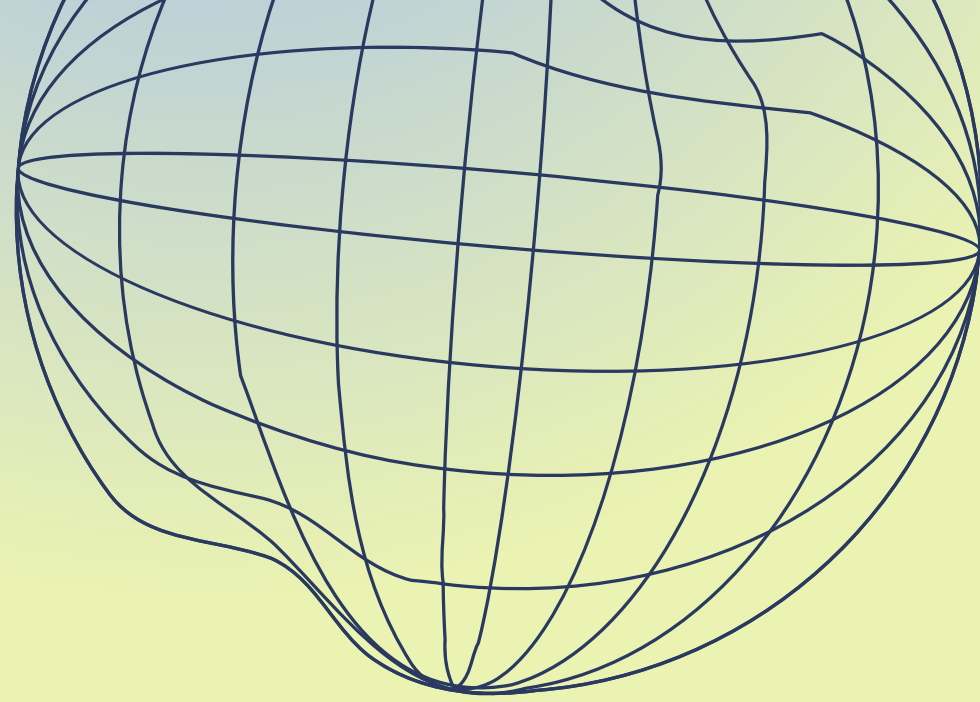
#groupproject

Objectives

The project aims to:

- ✓ Measure room temperature and humidity.
- ✓ Monitor soil moisture continuously.
- ✓ Automatically activate a water pump when the soil is dry.
- ✓ Prevent overwatering through threshold control.
- ✓ Detect environmental conditions that may lead to:
 - Mold growth
 - Root rot
 - Pest infestation
- ✓ Provide remote monitoring via IoT technology.





system architecture

/page 04

#groupproject

benefits



- Saves water
- Reduces human intervention
- Maintains healthy plant conditions

system workflow

01.



02.



03.



04.



05.



06.

Sensors collect environmental data every 5 minutes.

- Soil Moisture Sensor
- DHT11 Sensor

ESP32 processes the collected data.

The system evaluates:

- Soil moisture
- Temperature
- Humidity

If soil moisture falls below 35–40%:

- Relay activates
- Water pump starts watering

Watering continues until soil moisture reaches approximately 60%.

The pump automatically stops.

hardware components

#groupproject



hardware list

Components	Quantity
<u>ESP32</u>	1
<u>YL-69 Soil Moisture Sensor Module</u>	1
<u>DHT11 Sensor</u>	1
<u>5V Relay Module</u>	1
<u>5V Mini Water Pump</u>	1
<u>1m Water Pipe</u>	1
<u>Breadboard MB-102</u>	1
<u>Jumper Wires</u>	Several

DECISION-MAKING ALGORITHM

#groupproject

/page 06



Environmental Risk Assessment
Normal Condition (Low Risk)

Temperature:

- 18°C–30°C

Humidity:

- 40%–70%

Soil Moisture:

- 30%–60%

Result:

- Plant grows under healthy conditions.



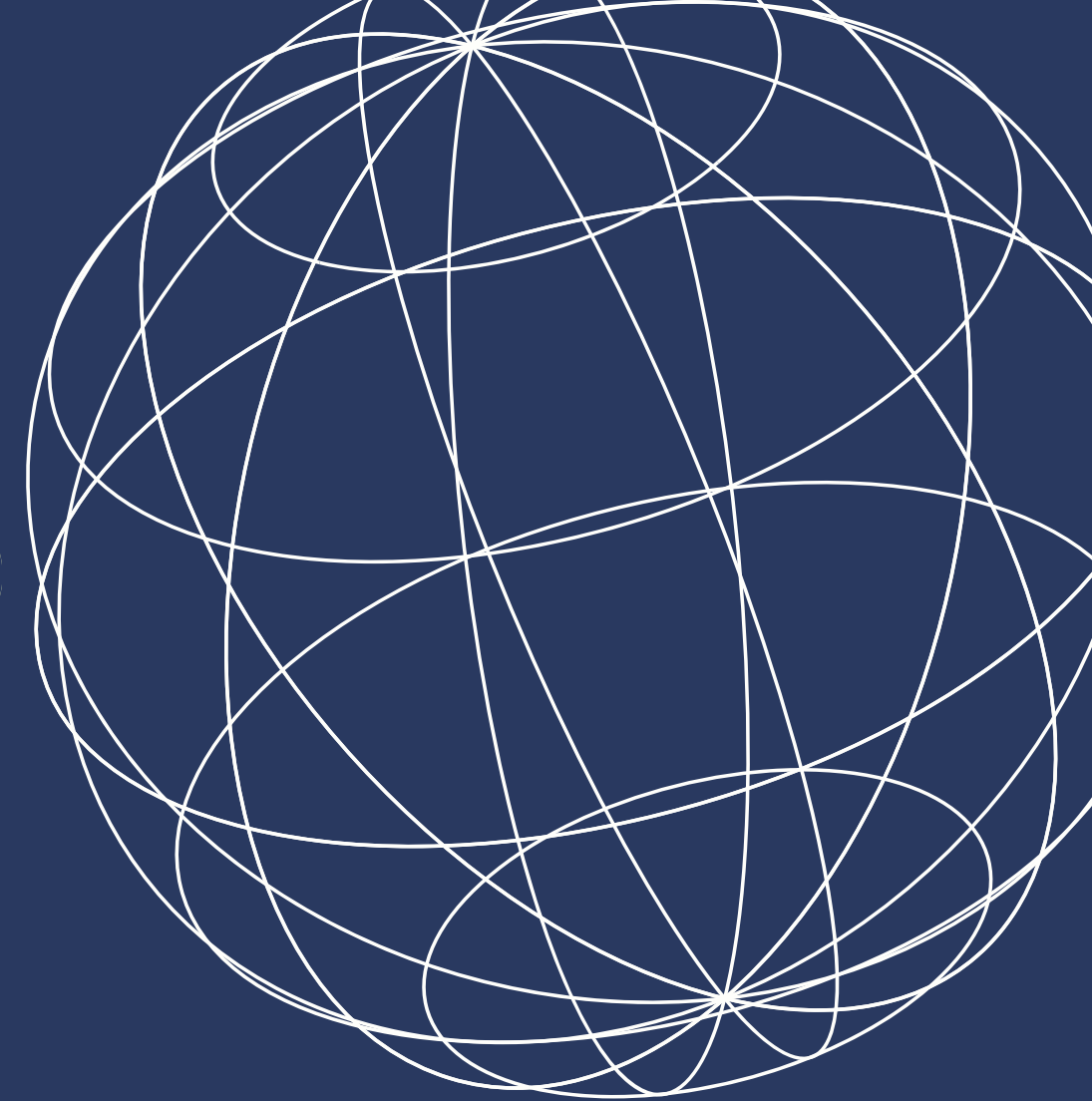
Mold Growth Risk (Medium Risk)

Conditions:

- Temperature: 20°C–30°C
- Humidity > 80%
- Soil Moisture > 70%

Result:

- Mold growth may occur.



Plant Health Monitoring

Normal Condition

<u>Parameter</u>	<u>Value</u>
Temperature	18–30°C
Humidity	40–70%
Soil Moisture	30–60%

Mold Growth Risk

Conditions:

- Temperature 20–30°C
- Humidity > 80%
- Soil Moisture > 70%

Result: Mold development may occur.

/page 07

environmental risk analysis

#groupproject



Disease & Pest Detection

/page 08

High-Risk Conditions

Root Rot Risk

Conditions:

- Temperature 22–30°C
- Humidity > 85%
- Soil Moisture > 80%

Symptoms:

- Yellow leaves
- Soft stems
- Dark roots

Pest Infestation Risk

Conditions:

- Temperature 25–32°C
- Humidity 70–90%
- Soil Moisture > 75%

Possible Pests:

- Mealybugs
- Fungus gnats



Software & IoT Platform

/page 09

#groupproject



Software Implementation

Development Environment

- Arduino IDE
- C++

Cloud Platforms

- ThingSpeak
- Blynk

Functions

ESP32:

- Reads sensors every 5 minutes.
- Controls relay.
- Uploads data to cloud.
- Generates warning notifications.
-

expected results

#groupproject

/page 10

System Performance

The system can:

- ✓ Monitor plant conditions in real time
- ✓ Automatically water plants
- ✓ Reduce risk of root rot
- ✓ Detect disease risks
- ✓ Provide cloud monitoring
- ✓ Store historical data

testing

#groupproject

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	A	B	C	D	E
1	created_at	entry_id	Temperature	Humidity	Soil Moisture
2	2026-06-21	61	31	80	2201
3	2026-06-21	62	31,3	78,3	2271
4	2026-06-21	63	31,3	78,3	2269
5	2026-06-21	64	31,3	78,3	2276
6	2026-06-21	65	31,3	78,2	2270
7	2026-06-21	66	31,3	78,2	2182
8	2026-06-21	67	31,3	78,1	2423
9	2026-06-21	68	31,2	78,2	2566
10	2026-06-21	69	31,2	78,2	2572
11	2026-06-21	70	31,2	78,2	2569
12	2026-06-21	71	31,3	78,3	2581
13	2026-06-21	72	31,3	78,4	2577
14	2026-06-21	73	31,3	78,8	2378
15	2026-06-21	74	31,4	78,8	2495
16	2026-06-21	75	31,4	78,6	2539
17	2026-06-21	76	31,4	78,5	2579
18	2026-06-21	77	31,4	78,4	1949
19	2026-06-21	78	31,5	78,3	2528
20	2026-06-21	79	31,5	78,1	2515
21	2026-06-21	80	31,5	78,1	2505

1	created_at	entry_id	Temperature	Humidity	Soil Moisture
2	2026-06-21	17	33,9	67,1	4095
3	2026-06-21	18	33,9	66,4	4095
4	2026-06-21	19	33,9	65,9	4095
5	2026-06-21	20	33,8	65,3	4095
6	2026-06-21	21	33,7	65,6	4095
7	2026-06-21	22	33,8	66,9	4095
8	2026-06-21	23	33,8	67,5	4095
9	2026-06-21	24	33,8	67,6	2730
10	2026-06-21	25	33,9	67,8	2841
11	2026-06-21	26	33,9	67,4	4095
12	2026-06-21	27	33,9	67,7	4095
13	2026-06-21	28	33,9	68,7	4095
14	2026-06-21	29	33,9	69,1	4095
15	2026-06-21	30	33,8	68,9	4095
16	2026-06-21	31	33,7	68,6	4095
17	2026-06-21	32	33,7	68,1	4095
18	2026-06-21	33	33,6	67,6	4095
19	2026-06-21	34	33,5	67,7	4095
20	2026-06-21	35	33,5	67,7	4095
21	2026-06-21	36	33,4	67,9	4095
22	2026-06-21	37	33,4	67,9	4095
23	2026-06-21	38	33,3	68,1	4095
24	2026-06-21	39	33,4	68,2	4095
25	2026-06-21	40	33,4	68,2	4095
26	2026-06-21	41	33,3	68,2	4095
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28	2026-06-21	43	33,3	68,5	4095
29	2026-06-21	44	33,3	68,4	4095
30	2026-06-21	45	33,3	68,4	4095
31	2026-06-21	46	33,3	68,4	4095
32	2026-06-21	47	33,3	68,4	4095
33	2026-06-21	48	33,3	68,5	4095
34	2026-06-21	49	33,3	69	4095
35	2026-06-21	50	33,3	69,1	4095
36	2026-06-21	51	33,3	69,1	4095
37	2026-06-21	52	33,3	68,9	4095
38	2026-06-21	53	33,3	68,8	4095
39	2026-06-21	54	33,2	69	4095
40	2026-06-21	55	33,2	69,2	4095
41	2026-06-21	56	33,2	69,4	4095
42	2026-06-21	57	33,1	69,4	4095

Conclusion

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#groupproject

The proposed system successfully combines:

- 🌱 Environmental Monitoring
- 🌱 Automated Irrigation
- 🌱 IoT Technology

Benefits:

- Reduced maintenance effort
- Improved plant health
- Water conservation
- Low implementation cost

Future Development

- Mobile App
- AI-based prediction
- Multiple plant support
- Solar-powered operation

Borcelle University

**thank
you.**

by: AE CÂY KHẾ TEAM

